

Fiefdom: the rules

Prototype v1 | Mechanics and adjustable balance values | September 30, 2026

Yes, this game is balanceable. Its main decision is how much reputation to lock in expansion while keeping enough free to protect existing land. Longer oaths, exposed borders and permanent fortifications create competing uses for the same currency. These rules make that decision measurable; the starting numbers still need playtesting.

The repeated loop

When	Player action	Result
Daily	Keep the goals in the ledger; inspect announced raids; order owned overlords to defend.	Earn reputation. Spend some to hold territory.
At weekly boundaries	Choose a frontier overlord and seal a land oath, or continue earning without expanding.	Lock a bond for 1, 2 or 4 weeks.
Whenever affordable	Improve a tile from I to II to III, repair it, or save a reserve.	Reduce future defense costs.
At oath settlement	Meet the signed terms.	Recover the bond, earn a fixed bonus and claim the land.
At season end	Finish a 46-week campaign.	Archive the realm and its achievements; begin a new season if desired.

Fixed mechanics; adjustable numbers

The map is fixed and only the camera rotates. The player starts on an invulnerable central tile, surrounded by three kingdoms that also fight each other. Each capturable tile has a unique overlord. Only connected frontier land can be claimed. Reputation can be **earned, locked or spent**; these are separate events. There is one active land oath. Tile strength has three tiers. No loss removes the home tile or ends the campaign.

Changes needed in the present ledger

Keep the Chronicle, seven-day contracts and Archive. Add a territorial oath over those weekly contracts. Cap funded duty slots at three; other duties remain personal tracking. Use a settled campaign purse rather than recalculating spendable money from editable history. Replace campaign wager multipliers with refundable bonds and fixed bonuses. Automatically settle game rewards; a weigh-in remains optional and does not decide success.

The current ledger pays +25 for closing even a poor week. In this proposal, that weekly bonus requires 70% completion, so automatic settlement cannot mint reputation from an empty ledger. Existing diary history and legacy wager totals are preserved as records; a new campaign starts with its own purse.

Map and campaign

A static realm, a permanent home and a finish date that does not depend on conquest.

Starting board

Use a radius-three hex map: **37 tiles**. One central home belongs to the player. Three outer capital tiles are invulnerable and belong permanently to their respective kingdoms. The remaining **33 tiles** begin divided into three connected sectors. Each capital must have a marked capturable royal-border tile reachable from home through bands 1, 2 and 3.

Band b	Position	Capturable tiles	Meaning
1	One edge from home	6	Nearby minor fiefs
2	Two edges from home	12	Marches
3	Three edges from home	15	Royal borderlands; 3 other outer tiles are capitals

Band is an authored, immutable risk value. On this starter map it equals shortest map distance from home. It never changes with ownership, routes or camera rotation. A tile is **supplied** if a path of player-owned tiles connects it to home. A frontier tile is enemy-owned and adjacent to supplied player land. Capitals are excluded from conquest and attacks.

State that must be stored

Object	Required state
Tile	ID, neighbors, band, owner, permanent overlord ID, tier, integrity, consecutive disconnected days; royal-border flag.
Player	Free reputation F, oath bond L, reserved defense O, retention priority, standing defense budget, active patrons.
Oath	Target overlord, start/end dates, copied terms and modifiers, bond, score snapshots, bonus-paid flag, pending-claim expiry.
Campaign	Start date, day index, fixed timezone, random seed, rule version, settled event IDs, first-pledge and territory history.

Time and ending

A campaign lasts **T = 322 daily turns**, or 46 weeks, approximately 10.6 months. Resolve once per day at 04:00 in the campaign timezone. A turn uses the ledger date on which it opens; that date can be corrected until the next 04:00 cutoff. Changing the computer timezone does not change the campaign timezone. Catch up absent days in order using saved standing orders.

The season continues even if its main achievement is reached early. A royal pledge is the first successful oath to a marked royal-border overlord, including a success whose claim is delayed. Aim to earn all three. Report average daily territory value, defined as the season mean of the sum of band* tier over supplied non-home player tiles, plus peak land and fulfilled oaths. There is no defeat outcome. A new season resets map and purse; diary and campaign history persist.

Reputation economy

Use the existing reward scale, with capped sources and explicit transactions.

Where reputation comes from

Source	Campaign reputation
A kept steps / calorie term	+10 / +10 for that day
Each funded duty kept	+4; at most H = 3 funded duties
Perfect day	+5 when both core terms and every funded duty are kept
Perfect streak	+1 per preceding consecutive perfect day, capped at +10
Weekly settlement	+25 if weighted weekly score is at least 70%; an extra +50 if flawless
First honored pledge to a tile overlord	Fixed band bonus B; paid once per overlord per campaign

Basic contracts are free and align with turns 1-7, 8-14 and so on. Each swears $h = 1$ to H funded duties. Contracts automatically renew, including while absent. Revise goals and duties at a weekly boundary unless a land oath freezes them. Goals remain player-selected. Weight change and more extreme health targets never multiply rewards.

```
day_income = 10*steps_kept + 10*calories_kept
+ 4*duties_kept + 5*perfect + streak_bonus
```

With three duties, a first perfect week earns **355** reputation. A mature perfect week earns **404**, including weekly bonuses. Adding a fourth duty cannot increase either ceiling. If fewer duties are funded, the ceiling falls. A week with no kept terms pays nothing.

The nurse must reconcile

```
F + L + O = initial_rep + settled_earnings + pledge_bonuses
- upgrades - repairs - upkeep - defense_wages
- bond_forfeitures
```

F is free to use; **L** is temporarily locked in the sole active oath; **O** is committed to the current defense. All stay nonnegative. Sealing moves reputation from F to L; reserving orders moves F to O. Returning a bond is a transfer, never a new earning. Wages are spent at battle settlement, even if a paid commander was not needed.

Holding more land has a cost

```
daily_upkeep = u * max(0, N - N_free)
defaults: u = 1 rep; N_free = 3 non-home tiles
```

At turn opening, fund owned tiles in the saved retention order: the first three are free, and each additional tile costs 1 reputation. Pay only what F can cover; there is no debt. Unfunded tiles lose one integrity at resolution. Home is always free. This creates a modest cost for a wide empire without granting passive income from land.

Every daily reward settles once into the campaign event ledger. Later diary edits update the diary but do not replay earnings, battles or purchases. Do not import lifetime ledger reputation or the old x2/x3 wager returns into this economy.

Land oaths

Long promises use the same weekly ledger; the bond makes expansion compete with defense.

Band	Duration D	Bond L	First bonus B	Score q	Category floor f
1	7 days	80 rep	28 rep	70%	60%
2	14 days	240 rep	84 rep	80%	70%
3	28 days	720 rep	252 rep	90%	80%

Seal and evaluate

An oath starts with a new weekly contract and covers 1, 2 or 4 consecutive contracts. Require F at least L, an eligible frontier target and enough season time for the oath to finish. Copy the chosen goals, funded duties and patron effects at sealing. These cannot change during the oath. Daily earnings and weekly bonuses continue; only the bond is locked.

```
s_d = 0.35*steps_kept + 0.35*calories_kept
+ 0.30*(funded_duties_kept / funded_duties_sworn)
S = sum(s_d) / D
```

Success requires **S at least q** and each category at least f: steps kept / D, calorie terms kept / D, and all funded duties kept / (h*D). Here h is the actual sworn duty count, from 1 to H. Duties form one capped pool, rather than a threshold per duty. Missing records count as missed terms. Use integer arithmetic; never round a below-threshold percentage into a pass.

Settlement and setbacks

Success: return the full bond to F. Pay B if this overlord has not already paid a pledge bonus this season, and mark that bonus paid even if the claim is delayed. Claim the target if it is still frontier land. Retain its current tier; reset integrity to its maximum. New land becomes usable next turn.

Failed or voluntarily canceled oath: return 90% of the bond, rounded down, and forfeit the rest. The default forfeiture rate is $\lambda = 10\%$. Pay no pledge bonus and gain no tile. Earned daily and weekly reputation remains. There is no contract burning that destroys health records.

If rival kingdoms exchange the target tile, the oath still concerns the same overlord. If access is lost, evaluate normally: success still returns the bond and pays the one-time bonus. Reserve a **7-day pending claim**; restore access and apply it after battles. Otherwise the claim expires. It does not block an oath to a different target; the same target cannot have two live claims or oaths. Pending claims also expire at season end.

Overlord patronage

Each overlord has a distinct authored trait using small template mixtures: for example bonus +5%, bond -5%, or forfeiture -2 percentage points. Equip at most **P = 2** supplied owned lords as patrons. Add effects; cap bonus and bond changes at 20%, and keep forfeiture at least 2%. Round bonds up and bonuses down. Signed effects stay fixed through settlement even if a patron tile is lost. Reclaiming a tile never pays its first bonus again.

Defense and fortification

A daily allocation decision, with permanent upgrades reducing the need for paid orders.

Tier	Base defense V	Upgrade cost	Role
I	6	Starting tier	Needs paid help for ordinary raids
II	14	160 rep: I to II	Holds nearby ordinary raids unaided
III	26	360 rep: II to III	Holds all ordinary raids unaided

Commands and combat

Each supplied owned tile provides its overlord as a commander; home provides the player as one commander. Give each commander one order per turn to defend its own tile or an adjacent supplied player tile, committing at most **40 reputation**. Several commanders can reinforce one tile. Reserve costs from F. Canceling or replacing an order before cutoff returns its unused O to F and frees the commander. Only final frozen orders become wages.

```
Defense_i = V[tier_i] + eta*sum(commander_rep_on_i)
eta = 0.5 defense per reputation
Minimum_rep_i = ceil(max(0, Attack_i - V[tier_i]) / eta)
```

If defense meets or exceeds the announced attack, the raid fails. Otherwise the tile loses one integrity. Each tile starts with **J = 3** integrity. There is no automatic healing. Repair one integrity for **48 reputation**, at most once per tile per turn. Zero integrity transfers the tile to its attacker. The new owner inherits one lower tier, with a minimum of I, and full integrity. Upgrades do not heal damage; a successful land claim does.

Band	Ordinary attack	Siege attack	Tier I / II / III rep needed for ordinary raid
1	12	18	12 / 0 / 0
2	18	24	24 / 8 / 0
3	24	30	36 / 20 / 0

An outer siege against tier III needs 8 reputation; against tier I it needs 48 and at least two commanders. Each raid is announced a full turn ahead with its exact target and strength. Upgrades and repairs do not alter the forecast. Upgrades are immediate, one tier at a time, on supplied owned tiles.

Standing orders and the portrait panel

The default policy funds upkeep, then reserves minimum sufficient defense in retention order within a discretionary **80-reputation daily cap**. It combines eligible commanders; choose own lord first, then neighbors in tile-ID order. Never overspend or auto-buy upgrades or oaths. If full defense is unaffordable, keep the money and show expected damage. Manual orders override the default without duplicating reservations.

Show the player portrait by default. Clicking a tile selects its overlord; clicking a capital selects that kingdom ruler. The panel has toggles for description and skill, plus the values relevant to decisions: trait, tier, integrity, supply, forecast and oath requirements. Small idle animation is cosmetic; turning the map never changes ranges or adjacency.

Kingdoms and daily order

Bounded rival actions, simultaneous battles and slow territorial decay.

The three enemies really fight one another

Each kingdom has one operation slot per turn and activates it with probability **a = 0.50**. It can raid only enemy land adjacent to its own capital-connected land. Home and all capitals are immune. Rival kingdoms use the same tier, integrity and defense comparison; their garrisons use base tier defense, without a simulated habit ledger or free extra operations.

```
Player_target_share = min(0.60, 0.20 + 0.02*N)
Siege_probability = 0.25; Siege_bonus = 6
```

If both player and rival targets exist, use that share to choose the player; otherwise choose a rival. With only player targets, use the same probability and otherwise rest; with no targets, rest. Pick by ruler preference: weakest tier, most damaged, or highest band. Seeded choices break ties. These are the rulers' visible tactical skills. Attack strength is the target's band value, plus the siege bonus if rolled.

Limit raids against the player to **C_attack = 2** per turn and one raid per target tile. Reroute conflicts or excess operations to another eligible rival target, or rest. Rulers remain present through their immune capitals. Generate and save forecasts from end-of-turn ownership for the next turn. Seeded choices are reproducible; reloading cannot reroll them.

Exact turn order

Step	Rule
Opening	Pay upkeep; mark unfunded tiles. Load forecasts; reserve affordable standing defense.
Player decisions	Log goals, seal oaths, buy upgrades/repairs and replace orders. Freeze at 04:00.
1. Earnings	Settle daily and weekly earnings once. New money cannot change frozen orders.
2. Battles	Use opening ownership/supply and final tiers/orders. Resolve all raids before transfers; spend wages and apply damage.
3. Territory	Apply decay and transfers; recompute supply. Lost lords stop serving next turn.
4. Oaths	Settle oaths and pending claims against the post-battle map. New land serves next turn.
5. Record	Save events, territory value and next forecasts; advance once. Archive after turn 322.

Supply decay and loss limit

Count disconnection from opening supply. A disconnected tile supplies no commander or patron. After **Q = 7** disconnected turns, lose one integrity each further disconnected turn. Reconnection resets the counter without healing. At zero integrity, decay or upkeep transfers the tile to the kingdom with the most adjacent tiles, otherwise to the nearest capital. Use fixed kingdom ID to break ties. Frozen raids and orders still resolve if their source tile is lost in that battle.

Allow at most **C_loss = 2** player tile transfers per turn across raids, upkeep and decay. Prioritize the highest-band zero-integrity tile, then lowest tile ID; defer other losses at one integrity. Home is excluded from every damage path. A prolonged absence can eventually leave only home, while its free contracts always provide a route back.

Tuning without redesign

Keep state transitions fixed. Change values in one versioned configuration between campaigns.

Adjustable parameter	Default
T; map size	322 turns; 37 tiles
H; daily reward values	3; 10/10/4/5; streak cap 10
D[b]	7 / 14 / 28 days
L[b]; B[b]	80/240/720; 28/84/252
q[b]; f[b]; lambda	.70/.80/.90; .60/.70/.80; .10
P; patron caps	2; 20%; forfeiture floor 2%
V[tier]; upgrade costs	6/14/26; 160/360
eta; commander limit	.50; 40 rep per lord
J; repair cost	3; 48 rep
a; siege chance; attack values	.50; .25; 12/18/24 +6 siege
N_free; u	3; 1 rep per extra tile/day
Q; pending claim window	7 days; 7 days
C_attack; C_loss; auto budget	2; 2; 80 rep/day

Useful ratios

```
Oath_cash_EV_per_day = [p*B - (1-p)*lambda*L] / D
Upgrade_payback_days = cost / (raid_rate * rep_saved_per_raid)
Reserve_target = expected_7_day_defense + 7*daily_upkeep
```

Here p is the measured chance of honoring that oath. Cash value excludes land, patron benefits and the cost of locking the bond. At the proposed values, B/D rises from 4 to 6 to 9 reputation per day. Higher-band oaths should offer stronger land and patron choices too; a larger total bonus alone is not enough.

Invariants to keep while changing values

D is a positive multiple of seven; keep T at 308-329 turns for the 10-11 month brief. Duration, bond and bonus rise by band; B/D also rises. Floors do not exceed score thresholds; both stay within 0 to 1. Targets are reachable. Home/capitals stay immune. Funds never go negative; payout IDs stay unique. Tier III holds ordinary raids; some sieges need help. Repair cost stays at least the maximum single-raid defense cost at tier I. Keep trait caps and exclude compounding purse rewards.

A complete sweep needs at least $6*7 + 12*14 + 15*28 = 630$ oath-days with one oath at a time. The 322-day campaign rewards a chosen realm and royal pledges. Three separate one-tile-per-band royal routes require **147 oath-days**, before saving, rebuilding or failed oaths; shared routes may reduce this.

Balance checks and next tests

Arithmetic and probability checks completed; the complete territorial game is not yet validated.

What the current numbers imply

Per-term adherence	Band 1 pass	Band 2 pass	Band 3 pass
85%: independent goals	85.59%	82.19%	7.48%
95%: independent goals	99.25%	99.91%	98.06%
85%: whole good/bad days	92.62%	64.79%	18.71%
95%: whole good/bad days	99.62%	96.99%	83.73%

Models use five terms per day: steps, calories and three duties, with the signed score and floors above. All table values use exact binomial enumeration, cross-checked with 100,000 seeded trials per adherence rate, band and model (seed 20260930). Days are independent in both models; whole-day results correlate the five terms. These assumptions do not predict any particular player.

Outer oaths are deliberately demanding: at 85% independent adherence, expected cash gain is -1.71 reputation per oath-day; at 95% it is +8.78. Bands 1 and 2 at 85% yield +3.26 and +4.63. A player should see the requirements before signing. If outer playtests produce frustration, lower q[3] first, rather than demand more extreme personal health targets.

Income and liquidity: independent 85% adherence earns about 33.28 reputation/day, including the gated weekly bonuses, before spending. From zero at home, three perfect days earn 114, enough for the first bond at a subsequent weekly boundary. Home and the first three fiefs have no upkeep. A failed oath returns 90% of its bond. Recovery requires keeping terms.

Fortification: on a band-3 ordinary raid, I to II saves 16 reputation, and II to III saves 20. At two such raids per week, their cash payback is approximately 35 and 63 days. Tier III in band 1 has no ordinary-raid savings over tier II; this is a reason to choose upgrades by exposure, rather than upgrade every tile indiscriminately.

How professionals approach balance

Set experience targets, model resource flows, simulate varied player behaviors, then watch real people play. Simulation exposes mechanical failures; player tests judge whether the decisions feel satisfying. Start with a small model and specific questions. Primary sources [1, 2] inform this approach.

Test next	Starting acceptance target - a design goal, not a result
Pacing and recovery	Typical player reaches 2-4 fiefs by week 4; after returning to home, rebuilds a first fief within 14 active days.
Strategic variety	Compare expansion, fortification and saving policies over 322 turns; no one policy leads in territory value across every adherence profile.
Pressure and usability	Daily decisions take roughly 2-5 minutes. No surprise raid, duplicate payout or negative purse. A missed day cannot erase more than two fiefs.
Long campaign	Simulate multiple seeds, 7/14/30-day absences and failed outer oaths. Measure income, upkeep, defense spend, locked share, tile churn and royal-pledge timing.

[1] *Machinations: How to Manage Complexity in Machinations* - small models and iteration.

[2] Volz, Rudolph & Naujoks: *Demonstrating the Feasibility of Automatic Game Balancing* (2016) - simulation and playtesting. All rules and defaults here are proposals. Repository basis: README and renderer lib/reputation.ts, contracts.ts, ledger.ts and types.ts.